[Outcome] Inventory | In Review

Outcome Name	Inventory
Product Manager	Lois Elliott
Key Stakeholders	Ryan Dice
Status	In Review
JIRA	
Figma	https://www.figma.com/design/INhQwAFGZOusl89pqQXq6W/Inventory-WIP_VA?node-id=39-592 Can't find link
Release Notes	

Product Vision

Target Audience

- IT Professionals - typically small to mid-sized IT teams and Managed Service Providers (MSPs) who need to discover, track, and manage the hardware and software assets across their environments. These range from solo IT admins managing a single office to MSPs managing dozens of client sites from a single account.

What their needs are

- Know exactly what hardware and software exist across my environment across Windows, macOS, and Linux without manually auditing devices or relying on spreadsheets.
- Quickly look up an asset's full profile (specs, warranty, owner, support history) when a user reports an issue, without having to log into the device or dig through multiple tools.
- Track the lifecycle of every asset from purchase to disposal - who owns it, where it is, its condition, and when its warranty expires, so I can plan replacements and manage budgets.
- Identify risk: assets out of warranty, devices not recently scanned, unapproved or unused software, and hardware nearing the end of life before they become incidents.
- Generate reports for finance, compliance, and management, such as asset valuations, software install counts, and warranty status, without exporting raw data and building reports manually.
- Manage multiple client sites from a single account (MSPs), with each client's assets clearly separated and mapped to the correct help desk organization.

Why we should do this

Inventory has been one of Spiceworks' foundational products since the on-premise era, and the current Cloud Inventory was deployed as an MVP that was never fully revisited. The result is a product that falls short of what IT Pros need: it's Windows-only, slow, and lacks capabilities that the on-premises version had and that competitors now offer as standard.

Community feedback consistently highlights three pain points: the product feels rigid (too many manual workarounds for basic tasks like bulk management and custom fields), the initial scan takes days, and platform support is incomplete (macOS and Linux agents are absent, leaving users with partial visibility).

Rebuilding Inventory addresses these gaps while creating strategic value. Asset- and software-level data is foundational to other Spiceworks products; it enriches the help desk with device context, enables smarter advertising and leadgen targeting through verified install data, and positions Spiceworks to expand into adjacent areas such as license management and security posture reporting. A modern, cross-platform Inventory also strengthens the overall product suite, giving users a reason to stay within the Spiceworks ecosystem rather than migrating to competitors.

Impact we expect to see

- Existing Cloud Inventory users migrate to the new product and see immediate value from cross-platform support, improved performance, and richer asset data.

- Net new user acquisition driven by a competitive, modern ITAM tool that addresses the gaps highlighted in community feedback and user interviews.
- MSP adoption driven by multi-site management from a single account - a capability the current product lacks entirely.
- Reduction in community feature request volume for core inventory capabilities (custom fields, bulk management, macOS/Linux support, ticket-to-asset linking).
- Increased Cloud Help Desk engagement through the CHD integration - technicians use asset context when triaging tickets, reducing resolution time.
- Inventory data used to improve the quality and relevance of vendor campaigns, driving higher-value leadgen through verified software install data and hardware profiles.

Glossary

Term	Simple Definition
ITAM	IT Asset Management: Managing gear from purchase to disposal.
SAM	Software Asset Management: Specifically tracking app licenses and costs.
Shadow IT	Unauthorized software or gear used without IT's knowledge.
CMDB	Configuration Management Database: Maps how all IT parts relate.
Endpoint	Any user-facing device (Laptops, Phones, Tablets).
Agentless	Scanning the network for gear without installing local software.
Agent-based	Using a small app on the device to "push" data (Best for WFH).
Probe / Collector	A local "scanner" that sends data from a branch office to the cloud.
API Siphoning	Pulling asset data directly from tools like Intune or Jamf.
SNMP	Simple Network Management Protocol

	The standard language for getting data from Printers and Switches.
OID	A specific data "address" inside a device (e.g., "Toner Level").
MAC Address	The unique hardware "SSN" for every network-connected device.
Provisioning	Setting up a new device with all the apps a user needs.
Harvesting	Taking back a software license from a user who isn't using it.
Zombie Asset	An asset that's "active" in the books but hasn't been seen in 30+ days.
ITAD	IT Asset Disposition: Securely wiping and recycling old hardware.
TCO	Total Cost of Ownership: The "real" cost of gear over its whole life.
Depreciation	How much value an asset loses every year for tax/budgeting purposes.
True-up	Comparing what you're using vs. what you paid for (Compliance check).
FinOps	IT and Finance working together to cut cloud and SaaS waste.
CyberITAM	Using inventory data as the foundation for your security strategy.
SSH	Secure shell. Could be used to remote into switches, linux/unix devices for gathering inventory or config management
WMI	Windows Management Instrumentation, used for inventory and control, often via powershell.
EoL/S	End of life/support

Microsoft Intune	Microsoft Intune—Endpoint Management | Microsoft Security (paid)
Jamf	Jamf Apple Device Management. Mac iPad iPhone TV Apple MDM (paid)
ABM	Apple Business Manager (free)
MDM	Mobile Device Management
UEM	Unified Endpoint Management
MSP	Managed Service Provider

Background Information / Discovery

Spiceworks Current and Legacy Inventory Products

Feature Matrix

Feature Matrix

	On Prem	Cloud Inventory
Identifying Devices		
Agentless Scanning	SNMP (simple network management protocol) - printers, routers, etc SSH (secure shell) - Linux, macOS WMI (Windows) - servers, desktop OS HTTP	IP Scanner does some of this functionality. You get MAC Address, IP and maybe open ports/manufacturer/OS.
Agent-based Scanning	Only Windows devices	Only Windows devices (servers/desktop OS) Linux and macOS agents did exist, but no longer work and have been removed

Reporting & Insights		
Hardware specs	Y	Y
Software / Applications installed	Y	Y
Device connection drill down to see what's connected to what	Y	N
Anti-virus reporting	Y	N
Warranty checker with alerts	Y	N
Low disk space alerts	Y	N
New stuff alerts & reports	Y	N
Mobile Device Management (MAAS360)	Y	N
Devices not recently scanned	Y	N
Device Alerting (eg, printer out of ink)	Y	N

IP Scanner

The IP Scanner agent provides an overview of devices on your network. Install it on one device to scan an IP range and retrieve details such as operating system, open ports, and IP addresses. This agent also acts as a Collection agent for its host device, gathering data for Inventory Online and Connectivity Dashboard. For more information on the IP Scanner, read this doc .

Feature Requests

Feature Requests

Topics tagged spiceworks-inventory-online-feature-request

Gemini generated summary

Feature Bucket	Concise Summary of Request	Associated Topic IDs

Integration & API	Users are seeking deeper connections between the inventory tool and external ecosystems (specifically Google Workspace/Chromebooks and Microsoft AD). There is also a strong demand for "closing the loop" by integrating ticket history directly into asset views and providing API access for custom integrations.	1245239, 1232876, 976874, 976151, 976686, 976520, 976523
UI Customization & Organization	This is a high-volume category. Users want the ability to create custom columns, rearrange data grids, and organize assets by "Client" or "Organization" (multi-tenancy). Requests also include custom grouping for locations and tab creation for specific asset attributes.	1177576, 1122915, 1099407, 1066467, 976875, 976775, 976150, 976519, 976600
Automation & Notifications	Focuses on moving from passive tracking to proactive monitoring. Users are requesting adjustable alerts and notifications for connectivity status changes, device health issues, and dashboard alerts.	976796, 976590, 976583, 976582, 976540, 976339, 976155, 976147, 976293
Enhanced Asset Data	Requests for more granular data points that the current tool does not capture natively, including laptop battery health, printer page	1245566, 976154, 976803, 976802, 1232876

	count/usage trending, and identification of unlicensed software.	
Platform Support & Import	Includes critical updates for the MacOS agent (specifically for newer versions like Sonoma) and the ability to perform mass imports for devices that cannot be scanned (like iPads, TVs, or Chromebooks) to maintain a complete manual inventory.	1113144, 1109056, 1094841, 1251750, 976519, 976155

Common Themes and Sentiment

- **The "Manual Burden" Sentiment:** A recurring theme in the feature requests (especially IDs 1094841 and 1177576) is that the current tool feels "rigid." Users feel they have to perform too many manual workarounds (like manual entry for thousands of devices or "sorting" by agent locations) because the tool lacks bulk-management features.
- **The "Disconnected Data" Theme:** Sentiment suggests the inventory tool is currently an "island." Users are frustrated that they cannot see helpdesk tickets next to assets (ID 1232876) or sync with the Google/Microsoft environments they already use (ID 1245239).
- **Platform Parity:** There is a notable sense of neglect among Mac users (IDs 1113144, 1109056) who feel the agent support lags behind the Windows version, impacting their ability to maintain a unified inventory.

Current Inventory Performance Challenges

Current Inventory Performance Challenges

From Craig

Why is performance poor?

- Scaling Issues: The architecture wasn't built for scale (TAT-25 and Mike McNaughton's comment). We appear to be sending full data scans every time rather than just changes (diffs). Lots of error logging is being sent.
- Throttling Loops: Traffic is always at peak and throttled. Agents retry blindly without prioritization (and limited randomization), creating a loop.

- Maybe Connectivity Dashboard :Has a high frequency of checks. Likely contributing to the congestion if it uses the same upload mechanism.

Why we haven't fixed it?

Once the product was no longer a company focus, tech debt reached a point where the app couldn't be built. Changes to personnel in 2024 removed the operations team and engineering from involvement in resolving the throttling problem.

Current Inventory Problems from a Dev Ops Perspective

From Tanay Apr 21, 2026

*As part of our ongoing **AWS cost optimization efforts**, we reviewed the **agent-registrar** application running in the agent-registrar namespace. Currently, there are approximately **90 pods deployed**, of which around **80 are provider pods** (the primary application pods responsible for handling traffic, excluding jobs and registration-related pods).*

*From our observations, there have been **no active deployments or code changes in the past two years**. Additionally, based on the attached logs, the application appears to be **continuously hitting the health-check endpoint** and subsequently **returning errors**, indicating repetitive health-check activity without meaningful workload.*

*Before proceeding with any changes, we would like to understand the **potential impact of reducing the provider pod count**. If we scale the pods down to roughly **20–30**, it should allow us to reduce the EKS worker node count by approximately **5–6 nodes**, resulting in an estimated **monthly cost saving of around $800–$900**.*

Reducing the pod count resulted in user complaints Spiceworks Device Inventory - no check ins?

Craig's recommendations for a rebuild

Craig's recommendations for a rebuild

- Dedicated Squad**:** Inventory changes constantly (new hardware, OSs, AV flags on our scanning approach). It cannot be deployed but not maintained. ET-611 shows that engineers had to update lists manually to recognize manufacturers and even software (Win 11 ticket in there somewhere). TAT-21 shows changes needed to our scan script; missing or incomplete data may cause problems when uploading. I'm not sure we have the logic to take only the good parts of an upload.
- Smart Architecture**:** Separate the "Registration" traffic (speed) from "Update" traffic (efficiency). If this is trivial, make a 3rd one just for error logs.

- Delta Syncs: Only upload changes, not full scans. - Option to force a re-sync to handle edge cases.
- Pull-Based Errors: Stop agents from auto-sending all errors; let us request them only when debugging is needed. Perhaps we could pick by group. Or a once-per-week snapshot from a selected number.
- Connectivity Dashboard. If we keep the connectivity dashboard, and it is a contributor, it should use a different upload method.
- MVP - This feels like it was deployed as a minimum viable product and never truly revisited (promises to bring feature parity with the desktop app). It didn't help IT pros as much as our prior app, so it didn't gain as much traction, so we didn't dedicate resources, etc.
- Consider osquery or something like it - let's use open source (I know updating and dependencies), but this removes us needing to figure out how to scan stuff and keep it up to date. osquery is a scanner that works on Windows, Linux, and macOS. (Post engineeering POC it was determined that osquery did not collect all the required data)

Competitors

See Competitors Inventory Systems

Technical Considerations

See Technical Considerations Gemini Discovery

Scope and High-Level Features

These requirements are not maintained beyond the requirements phase.

Feature Roadmap Matrix

	Current cloud inventory	Now (MVP)	Next	Later	Maybe
Asset Discovery	IP Scanning agent (basic) Collection agents for gathering detailed asset data	Scanning and collection agents extended to macOS and Linux	Support for mobile devices		Automatically pull end-dates from Dell/HP/Lenovo using serial numbers

	Windows only Multi-site support via multiple scanning agents	Agentless network scanning via SNMP (printers, routers, switches) Multi-site support without multiple accounts (MSP)			Docker or Kubernetes pod information for Linux servers
Hardware Data • Minimum data list Inventory Online Scan Details - Spiceworks	Hardware and OS specs collected and displayed (Inventory Data Fields) First seen and last scanned dates		Health data (laptop battery health, disk health) Security features and status (BitLocker, FileVault, antivirus) on Windows, macOS, and Linux		Identify which assets are connected to each other
Software Data • Minimum data list Inventory Online Scan Details - Spiceworks	Software installed across all assets, with install counts		Subscription Management License Management Compliance Management		

User Data	Users who have logged on to a device				

Asset Metadata & Configuration		Ownership assignment Location assignment Lifecycle status (Active, In Storage, Under Repair, Retired, Disposed, Lost/Stolen) Purchase information (date, price, PO number, vendor) Warranty information (provider, start/end dates) Custom fields (text, dropdown, number, date) Manage the list of asset owners Manage the list of locations	Asset dashboards - by device type, by location Location pictures	Asset label creation Asset QR codes Barcode scanning	
Asset Management	View asset details	Bulk update of manual and custom fields		Manually upload other “device types”	Office Map / site map with asset overlay

	Edit asset data Delete assets Manual asset entry	Manual notes on an asset timeline Asset timeline/audit history		(desks, phones) Contact Management Ability to pin a note about an asset Bulk import rather than individual manual asset entry	Device locator User checkin / checkout process including signed receipts from recipients
Search, Filtering & Saved Filters	Hardware asset search Software asset search	Save search criteria as Saved Filters			
Reporting & Exports	Export data - hardware and software (CSV / XLSX) Default filters: All hardware, Devices with agents, All software, Scanning agents	Utilize Saved Filters as Reports	Schedule Reports Connectivity Dashboard (replacement) Set up alerts if a particular asset goes offline	Set up rules to tag assets (e.g., "Obsolete" or "Critical" based on age > 5 years) Set up alerts (low disk space, low toner, out-of-warranty, new asset)	Predictive dashboards/reports (e.g., assets most likely to fail or need replacing in 6 months)

Integration with CHD	Create tickets from inventory for a specific asset Link inventory assets to a CHD ticket (from CHD) View all tickets associated with an asset	CHD organization mapping for multi-site MSPs			Cross application user management
Integrations				InTune Google Workspace Admin Console	Expose the data for AI consumption
User Access & Account Management	Manage who has access to the inventory	SSO			

MVP High-Level Requirements

The MVP for Inventory must include the functionality of our current Cloud Inventory product, along with additional functionality, to be useful to users.

For review

- Blue - Design
- Green - Ryan
- Yellow - Srikar
- Purple - Craig

Definitions

- Scanning Agent - Scans a defined IP range on the network to discover devices.
- Collection Agent - Installed on a device and collects the data about that device only.
- Device Types - (e.g., Laptop, Printer, Server, Switch). The device type list is fixed
- Discovered field - (e.g., OS Version, RAM, MAC Address). Automatically updated and overwritten upon every agent check-in. Not editable unless the asset was manually added.
- Calculated / Derived field - (e.g., Last check-in date). Automatically updated and overwritten upon every agent check-in.
- Built-in Manual fields - (e.g., Lifecycle Status, Warranty Dates, Owner, Location) *Never* modified by the agent payload. Updates to these fields can only occur via the Web UI, Bulk Updates
- Custom fields - Additional fields that users may add to their accounts to manually append additional information.

Assumptions

- The data model must support querying in both directions: all software on a device, and all devices with a specific software title + version.
- Specific field information is not included in this document; refer to Inventory Data Fields
- Standard Behavior for Grid Data: Unless specified, any grid data should include sortable, resizable, and re-orderable columns, as well as suitable pagination.
  - Column widths should persist per user across sessions.
  - Double-clicking a column border should auto-fit the column to its widest visible content.
  - A suitable empty state should be displayed
- Standard Behavior GA click tracking: Unique class names added to all buttons and links to enable GA click tracking using the agreed naming convention
- Standard Behavior Datadog logs: Relevant Datadog logs added for all actions / API calls with the correct level to aid troubleshooting and debugging.
- The current Cloud Inventory Scanner and Collection Agent approach is suitable if the multi-site and multi-organization MSP requirement is met, and performance improves. This document omits full requirements; review existing applications.
- There is no user role management for Inventory; the use of IT Pro, IT Manager, and IT Technician in user stories is intended to identify the different personas who would use Inventory. All users who have access can perform all actions.

Asset Discovery

The MVP must replicate and extend Cloud Inventory's current discovery capabilities, with a focus on cross-platform support and improved performance.

User Stories

- As an IT Pro, I want a scanning agent that supports Windows, macOS, and Linux, so that I can discover assets across all major operating systems in my environment.
  - The agent should support automatic updates without requiring manual reinstallation
  - Specific operating systems versions TBC
- As an IT Pro, I want to configure the IP ranges or subnets that my scanning agent should scan, so that I can control which parts of my network are discovered.
  - Users can define one or more IP ranges or subnets per scanning agent.
  - Users can exclude specific IP addresses or IP address ranges from scanning.
- As an IT Pro, I want collection agents that I can install on Windows, macOS, and Linux devices to gather detailed asset information, so that hardware specs, OS details, and installed software are automatically reported to the inventory.
  - The agent should support automatic updates without requiring manual reinstallation
- As an IT Pro, I want agentless network scanning via SNMP for network-connected devices (e.g., printers, routers, switches), so I can discover and track devices that cannot run agents.
  - SNMPv3 should be the default. SNMPv2c may be supported, but should not be the recommended option.
  - Users should be able to upload custom SNMP MIB files so that new device types can be inventoried without a product update.
- As an IT Pro, I want to download, install, and manage scanning and collection agents from within the Inventory application, so that I can deploy and maintain agents across my environment.
  - Users can download agent installers for each supported OS from the application.
    - Includes a silent/mass deployment option (e.g., MSI for Windows, pkg for macOS)
  - Users can view a list of all registered agents and their status (online, offline, last check-in).
  - Users can decommission/remove an agent
- As an IT Pro, I want to manage multiple sites belonging to the same organization from a single account, so that I don’t need to create and maintain separate accounts for this separation.
  - Replaces the current model, where multi-site requires multiple scanning agents tied to separate accounts.

    - This is not the same as the MSP use case below; it is a single organization that operates across multiple sites.
    - Each site's assets should be distinguishable within the single tenancy (e.g., by organization or site label).
    - The user must be able to create, rename, and delete sites from within the application
- As an MSP, I want to manage inventory across multiple organization/client sites from a single account, so that I don't need to create and maintain separate accounts for each organization/client.
    - Replaces the current model, where multi-site requires multiple scanning agents tied to separate accounts.
    - Each organization's assets should be distinguishable within the single tenancy (e.g., by organization label)
    - The user must be able to create, rename, and delete organizations from within the application
- As an IT Pro, I want the system to automatically flag potential duplicate assets when they are discovered, so that I can keep my inventory clean and avoid counting the same device twice.
    - After each agent check-in or manual asset creation, the system should compare the new/updated asset against existing assets to identify potential duplicates. The system should never auto-merge or auto-delete.
    - Common scenarios that generate duplicates:
        - A device is manually added, then later discovered by an agent.
        - A device is reimaged or renamed, causing the agent to register it as a new asset.
        - Multiple scanning agents on different subnets discover the same device.
- As an IT Pro, I want the option to force a scanner to do a rescan if I know something has changed on the network.
- As an IT Pro, I want the option to force a collection agent to refresh the data on a specific device if I know something has changed on that device.

Hardware Asset Data

Discovered Data

User Stories

- As an IT Pro, I want the collection agent to discover and display core device information so I can build a complete hardware profile for each asset.
    - The full list of discovered fields is defined in the [Asset Data Spreadsheet](#).

  - These fields are automatically updated on every agent check-in and are not user-editable (unless the asset was manually added).
- As an IT Pro, I want to see disk and partition information for each asset, so that I can monitor storage capacity and identify devices running low on space.
  - A device may have multiple disks and partitions
- As an IT Pro, I want to see network adapter details for each asset so I can identify IP addresses, MAC addresses, and network configurations.
  - A device may have multiple network adapters
- As an IT Pro, I want to see which peripherals are connected to each asset so I can track monitors, keyboards, printers, and other attached devices.
  - A device may have multiple peripherals
- As an IT Pro, I want to see video controller information for each asset, so that I can identify GPU hardware and driver versions.
- As an IT Pro, I want the system to automatically update hardware records when a previously discovered item is no longer detected on a device, so that my hardware inventory reflects what is currently installed and I can track changes over time.
  - Handles situations where a hard disk is replaced or a peripheral is removed.

Calculated / Derived fields

User Stories

- As an IT Pro, I want to see when an asset was first discovered and when it was last scanned, so that I can identify new devices and spot assets that haven't checked in recently.
  - "First Seen" is the date the asset first appeared in the inventory.
  - "Last Scanned" is the date of the most recent agent or scanner check-in.
  - Automatically maintained and not user-editable.

Software Data

User Stories

- As an IT Pro, I want the collection agent to discover and report all software installed on each device, so that I have a complete picture of what's running across my environment.
  - Software is logged exactly as written in the source: Windows Registry (Uninstall String), macOS Applications folder, or Linux package manager.

    - Each unique combination of Name + Version is tracked as a separate entry (e.g., *Google Chrome v120.0.1* and *Google Chrome v120.0.2* are distinct records).
    - Software data is automatically updated on every agent check-in.
    - The full list of discovered fields is defined in the Asset Data Spreadsheet.
- As an IT Pro, I want the system to automatically update software records when a previously discovered application is no longer detected on a device, so that my software inventory reflects what is currently installed and I can track changes over time.
- As an IT Pro, I want the system to calculate an Install Count for each software title and version, so I can see how widely each application is deployed.
    - Install Count = the number of devices with that exact Name + Version combination currently installed.
        - This is a calculated/aggregated field, not discovered per device.

User Data

User Stories

- As an IT Pro, I want the collection agent to discover and report user accounts and login activity for each device, so I know who is using each asset and when.
    - The agent should capture all user accounts that have logged into the device, not just the currently active session.
    - User login data is automatically updated on every agent check-in.
- As an IT Pro, I want the system to track when a user stops appearing in device login records, so I can identify stale accounts and understand device usage patterns.

Asset Metadata & Configuration

Built-in Manual Fields

User Stories

- As an IT Pro, I want to assign an Owner to an asset from a configurable dropdown, so that I can track who is responsible for each asset.
    - Users can define their owner list via configuration settings (Name, Email Address).
    - Owners can be scoped globally or restricted to a specific organization.
    - Default to blank on first discovery

- As an IT Pro, I want to assign a Location to an asset from a configurable dropdown to track its physical location.
  - Users can define their location list via configuration settings.
  - Locations can be scoped globally or restricted to a specific organization.
  - Default to blank on first discovery
- As an IT Pro, I want to set an asset's Lifecycle Status to track its lifecycle.
  - Dropdown options: `Active`, `In Storage`, `Under Repair`, `Retired`, `Disposed`, `Lost/Stolen`.
  - The lifecycle status values are fixed and cannot be configured.
  - Lifecycle status transitions are free-form (any status to any status)
  - Default to Active on first discovery
- As an IT Manager, I want to record asset purchase information to track costs and procurement details.
  - Purchase Date (Date Picker)
  - Purchase Price (Numeric with currency)
  - Purchase Order (PO) Number (Alphanumeric)
  - Vendor / Reseller (Text)
  - Default all fields to blank on first discovery
- As an IT Manager, I want to record warranty information for an asset to track coverage and expiration dates.
  - Warranty Provider (Text)
  - Warranty Start Date (Date Picker)
  - Warranty End Date (Date Picker)
  - Warranty Notes (Text Field)
  - Default all fields to blank on first discovery

Custom Fields

User Stories

- As an IT Pro, I want to create custom metadata fields for assets to capture data specific to my organization's requirements.
  - Users can create, edit names/options, delete, or drag and drop to reorder how custom fields appear in the asset detail pane.
    - Deletion is permanent, and all data stored in this field across all assets will be removed.

  - Supported data types: Single-line Text, Multi-line Text, Dropdown Selection, Number, Date.
  - Custom fields can be scoped globally to "All Assets", restricted to specific Device Types (e.g., a "Toner Model" field visible only on Printers), or restricted to specific organizations.
  - Custom fields are strictly manual and will never be populated or overwritten by agent scans.
  - No maximum number of custom fields per account
  - No maximum number of options in a dropdown custom field
  - Custom fields cannot be marked as required

Configuration

User Stories

- As an IT Pro, I want a configuration interface to create, edit, and delete entries in my Owner list, so the Owner dropdown on assets stays up to date as staff join and leave.
  - Each owner entry consists of: Name, Email Address (at minimum)
  - Owners can be scoped globally or restricted to a specific organization.
  - Deleting an owner who is currently assigned to one or more assets should prompt a warning showing how many assets are affected, and either reassign them or leave the field blank.
  - Owners cannot be imported in bulk
- As an IT Pro, I want a configuration interface to create, edit, and delete entries in my Location list, so that the Location dropdown on assets reflects my actual sites and facilities.
  - Each location entry consists of: Name, Description (at minimum).
  - Locations can be scoped globally or restricted to a specific organization.
  - Deleting a location currently assigned to one or more assets should prompt a warning indicating how many assets are affected.

Asset Management

Manually Adding Assets

User Stories

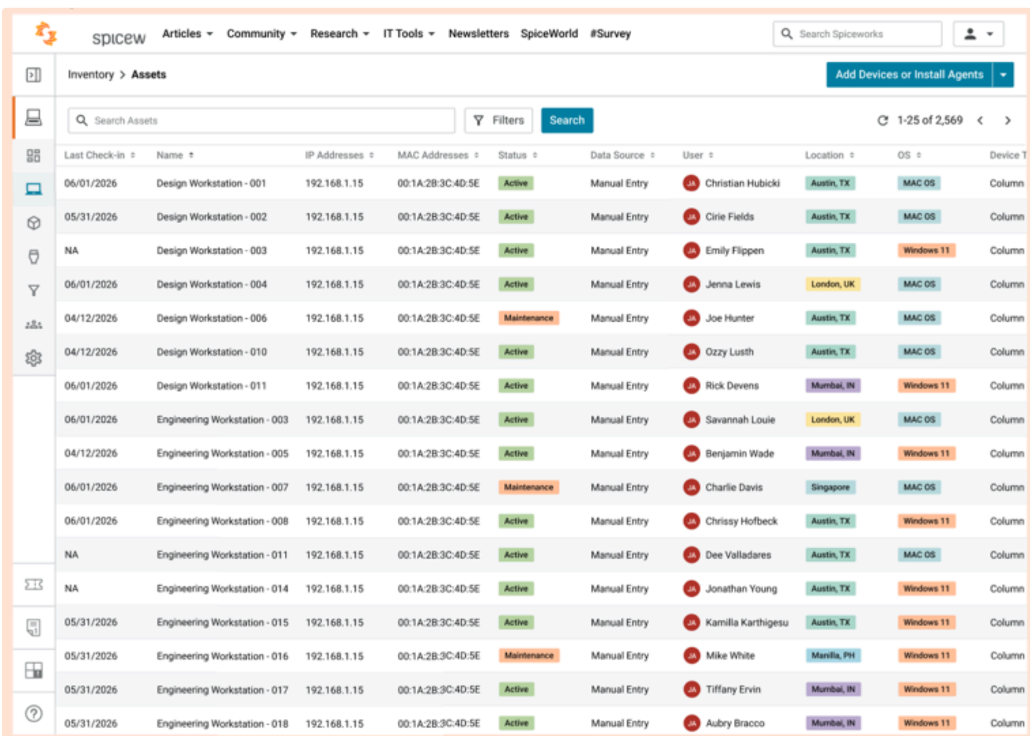
- As an IT Pro, I want to manually create an asset record for devices that do not support agents or SNMP (e.g., a projector or offline server), so that I can maintain a complete inventory of all equipment.

- Required Fields
  - Name (device name- the primary identifier in the grid)
  - Device Type (dropdown - e.g., Desktop, Laptop, Server, Printer, Network Switch, Other)
- The full list of required and optional fields is defined in the [Asset Data Spreadsheet]
- After successful creation, the user should be taken to the new asset's detail view, where they can continue adding information.

- As an IT Pro, I want the system to automatically flag a manually added asset as a potential duplicate asset if an agent later reports from the same device, so that I don't end up with duplicate records.

Hardware Asset Grid

The Hardware Asset Grid is the foundational interface for viewing and managing hardware inventory.



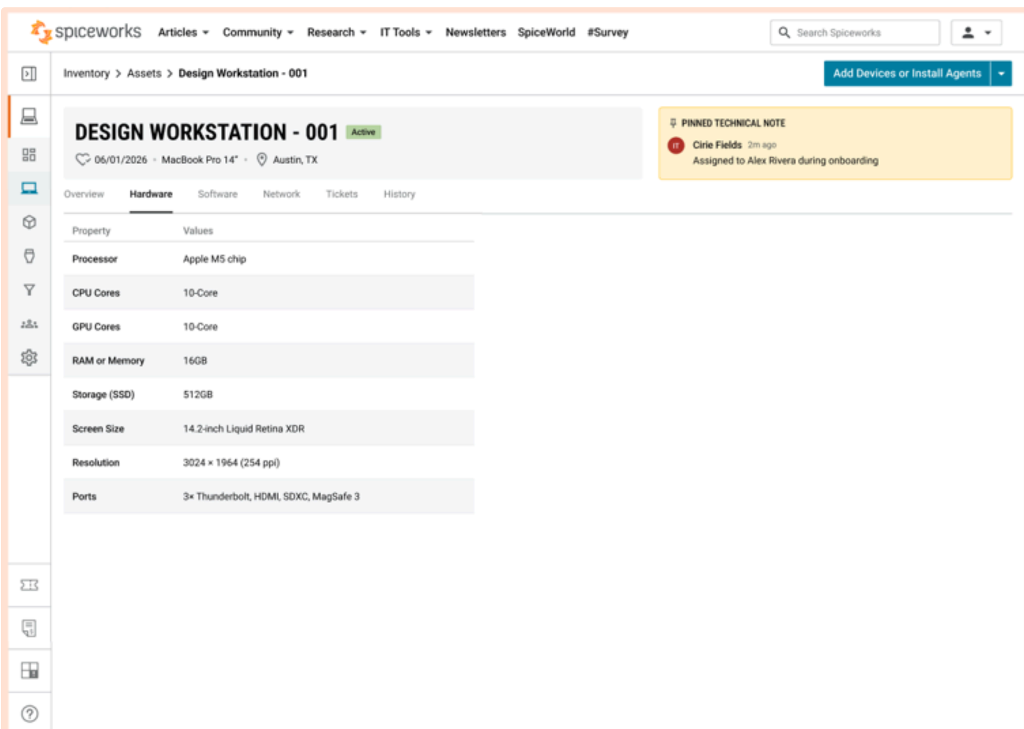
Concept Design

User Stories

- As an IT Pro, I want a primary asset grid that displays all my hardware inventory, providing a central interface for viewing and managing assets.
  - The default sort order is most recently scanned assets first

- As an IT Pro, I want a column picker that lets me show or hide available fields in the Hardware Asset Grid, so that I can customize the view to my needs.
  - The column picker should list all single-value discovered, calculated, manual, and custom fields.
  - Column visibility preferences should persist per user across sessions.
  - Fields that can be used in the Hardware Asset Grid are defined in the [Asset Data Spreadsheet]
- As an IT Pro, I want the Hardware Asset Grid to indicate when new or updated data is available, or to automatically refresh since I last loaded the view, so I'm always working with the latest information.
- As an IT Pro, I want to easily tell which assets in the Hardware Asset Grid were manually added vs discovered by an agent, so that I know which records are maintained by automated scans and which depend on manual updates.

Viewing Assets



Concept Design

User Stories

- As an IT Pro, I want to click on an asset in the grid to view its full details, so that I can quickly access all information about a specific asset.

    - Asset information is divided into different tabs to keep data organized with a persistent header
    - The tab the user last viewed should be remembered per session, so navigating back to an asset returns to the same tab.
    - Tabs that contain multiple records (Storage, Network, Peripherals, Video Controllers, Users, Software) to display data in a grid layout with one row per item, so that I can scan and compare multiple entries easily. Pagination is not required
    - The tabs and fields on each tab are defined in the Asset Data Spreadsheet
- As an IT Pro, I want a clear way to navigate back to the Hardware Asset Grid from the Asset Detail View, preserving my previous grid state (sort order, filters, scroll position, page), so that I don't lose my context.
    - A "Back to Assets" link or breadcrumb should be visible at the top of the Asset Detail View.
    - Returning to the grid should restore the user's previous sort, filters, selected page, and scroll position.

Editing Assets

User Stories

- As an IT Pro, I want to edit an asset's details to correct or update information as needed.
    - Only Manual and Custom fields can be edited
    - For manually added assets, all fields available on creation are editable
- As an IT Pro, I want every manual edit to an asset's fields to be recorded in the asset's audit history, so that I can see who changed what and when.
    - Each edit should log: field name, old value, new value, user who made the change, and timestamp.

Deleting Assets

User Stories

- As an IT Pro, I want to delete an asset from the Asset Grid or the individual asset view to remove obsolete or incorrect records.
    - Before deleting a discovered asset, a confirmation modal warns: *"You are about to delete X selected asset(s). If these assets are still active on your network and have agents installed, they will automatically reappear in this list upon the next scheduled agent scan, and their historical ticket association in Cloud Help Desk will be fully preserved."*

    - Before deleting a manually added asset, prompt the user to confirm they want to delete the asset.
    - Out of Scope
        - Bulk Delete

Merging Assets

User Stories

- As an IT Pro, I want to merge duplicate assets into a single record to consolidate their data and maintain an accurate inventory count.

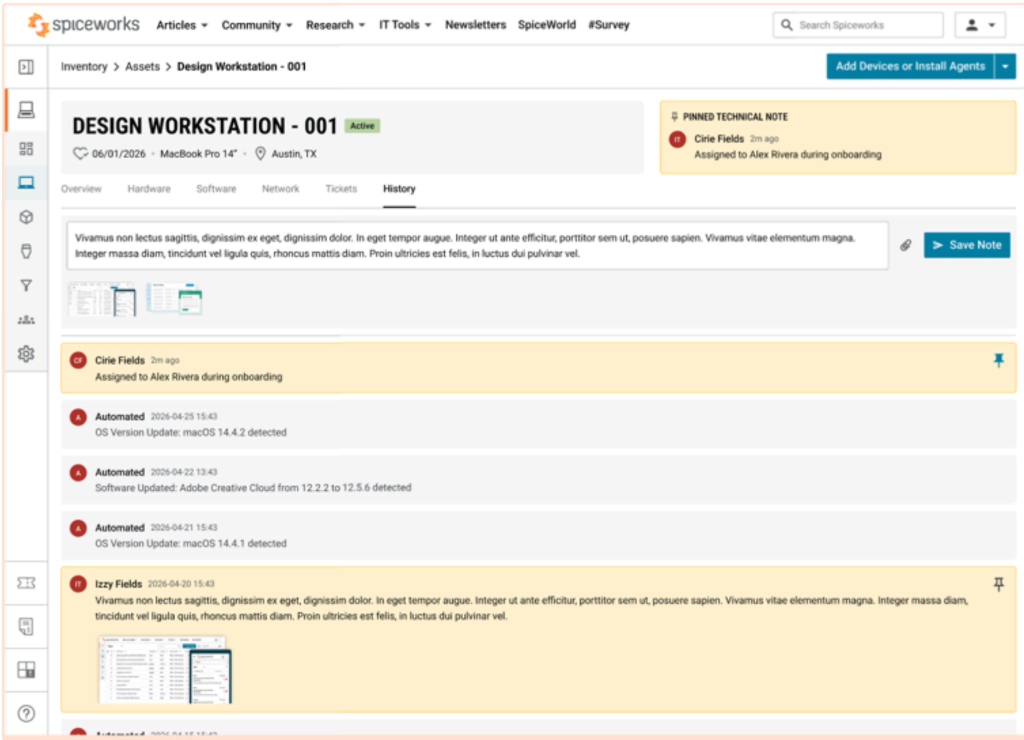
Bulk Update

User Stories

- As an IT Pro, I want to bulk-update manual and custom metadata fields across multiple selected assets simultaneously, so that I can efficiently manage large numbers of assets without editing them one by one.
    - Bulk update is enabled when the user checks 1 or more asset boxes in the Hardware Asset Grid. Only 1 page can be selected at a time.
    - Only fields classified as Manual or Custom Fields are eligible. Automatically scanned or system-calculated attributes are not available.
    - Each field should default to "Keep as is" - the user must explicitly set a new value for it to be included in the update.
    - Only the fields the user explicitly modifies will be updated; unmodified fields must retain their existing individual values (e.g., bulk-updating "Location" must not change "Owner" or "Warranty End Date").
    - If a custom field isn't applicable to the asset's device type, it should be skipped.
- As an IT Pro, I want to see the progress of a bulk update and be notified of any failures, so that I know the operation completed successfully or can take action on errors.
    - A progress indicator or notification when complete.
    - A summary of results: X assets updated successfully, Y assets failed.
    - For failures, a list of affected assets and the reason (e.g., "Asset XYZ: field validation error").

Historical Audit Trail & Manual Notes

Maintains historical logs of automated and administrative interventions for security, reporting, and lifecycle audits.



Concept Design

Asset Notes

User Stories

- As an IT Technician, I want to submit free-text manual notes on an asset's timeline (e.g., "RAM upgraded to 32GB on 10/12/2026 by John") to document maintenance, changes, and observations.
  - Notes cannot be deleted or modified once saved.
  - Notes can use markdown formatting
  - Notes should support file attachments
- As an IT Manager, I want to see exactly who wrote each note, so that I can follow up with the right technician if I have questions about a maintenance action.
  - Each note must display the full name of the user who created it
  - If a user's account is later removed, their name should still appear on historical notes (not replaced with "Deleted User" or similar).

Audit History

User Stories

- As an IT Pro, I want the system to automatically record significant changes to an asset's data in an immutable audit log, so that I can see exactly what changed and trace accountability.
  - The audit history is retained indefinitely
  - Each audit entry must record:
    - Field name (e.g., "Location", "Lifecycle Status", "Warranty End Date")
    - Old value (the value before the change; blank if the field was previously empty)
    - New value (the value after the change; blank if the field was cleared)
    - User (the person who made the change - or "System" for agent-driven updates)
    - Timestamp
    - Change source (manual edit, bulk update, agent check-in, or system-calculated)
  - For bulk updates, each affected asset's audit log should contain its own entry with the specific old/new values for *that* asset (not just "bulk update applied").
- As an IT Pro, I want the audit log to capture all meaningful events in an asset's lifecycle, so that I have a complete and trustworthy history.
  - Agent check-ins where nothing changed (routine heartbeats) do not need to be logged

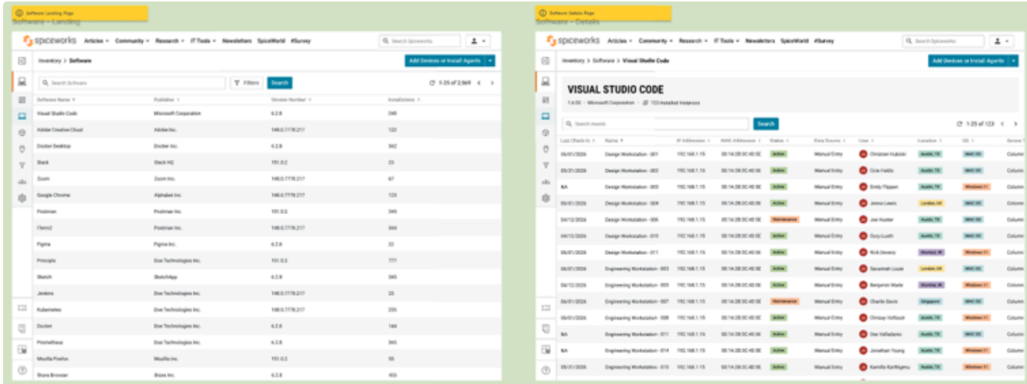
Asset Timeline

User Stories

- As an IT Pro, I want manual notes and automated audit entries for an asset to appear together on a single chronological asset timeline, so that I have one place to see the complete history of an asset.
  - Users can view Notes and audit history entries for an individual asset via a dedicated "Audit History" tab in the Asset View.
  - The timeline should load an initial set of entries (e.g., the most recent 50) and support "Load more" or infinite scroll for older entries.
  - Notes should be visually distinct from system-generated audit entries

Software Asset Grid

The Software Asset Grid displays an aggregated view of all software discovered across the entire organization.



Concept Design

User Stories

- As an IT Pro, I want an aggregated Software Asset Grid that displays all discovered software across my entire organization, so I can see what's installed and how widely each application is deployed.
  - Default sort on first load: Install Count descending.
    - If Install Count is tied, secondary sort by Software Name ascending (alphabetical)
  - First Discovered - the earliest date this software title + version was first seen on any device in the organization.
  - Last Discovered - the most recent date this software title + version was detected on any device (useful for spotting software that's no longer being installed).
- As an IT Pro, I want a column picker in the Software Asset Grid that lets me show or hide additional columns beyond the defaults, so that I can customize the view to my needs.
- As an IT Pro, I want the Software Asset Grid to show only currently installed software by default, with an option to include removed/inactive software, so that the grid reflects the current state of my environment without clutter.
- As an IT Pro, I want the drill-down from a Software Asset Grid row to show me all hardware assets with that specific software title and version installed, displayed in a grid with the same interaction patterns as the Hardware Asset Grid, so that I can investigate installations at the device level.

Search, Filtering & Saved Filters

Searching & Filtering Hardware Assets

- As an IT Pro, I want to be able to search across hardware assets to quickly find assets by keyword.
  - A single search bar at the top of the grid searches across designated text-based fields.

  - Fields that can be used in text search are defined in the [Asset Data Spreadsheet]
  - Text search supports single, multi, and exact phrases and wildcards functionality
    - Single Words: This will match on the complete word. If you want to do a partial search, you will need to use wildcards.
    - Multiple Words: Use AND (e.g., `WordA AND WordB`) to find assets containing all terms, or OR to find one or the other.
    - Exact Phrases: Wrap your search in double quotes (e.g., `"search phrase"`).
    - Wildcards: Use * for partial matches. For example, `abc*` finds words starting with those letters, while `*abc*` finds words containing them, and `*abc` finds words ending in those letters.
    - Smart Matching: The system uses “stemming” to find related words. Searching for “crash” will automatically include results for “crashing” or “crashed”.
    - Case Sensitivity: Search is generally case-insensitive but operators like AND and OR must be capitalized.
- As an IT Pro, I want to add filters to narrow the hardware asset list so I can build more precise queries.
  - Hardware data fields that can be used as Filters are defined in the [Asset Data Spreadsheet]
  - Date-related fields support filtering by relative dates and custom date ranges.
  - Additional filter options
    - Custom Fields
    - Scanner
    - Manually Added
    - Organization
    - Specific software installed
- As an IT Pro, I want to be able to bookmark a specific filter so that I can reopen it later
  - This is different from Saved Filters.

Searching & Filtering Software Assets

The Software Grid features a search and filter system that functions in a manner identical to the Hardware search and filter interface

User Stories

- As an IT Pro, I want to be able to search across software assets so I can quickly find them by keyword.

- A single search bar at the top of the grid searches across designated text-based fields.
- Fields that can be used in text search are defined in the Asset Data Spreadsheet
- Text search supports single, multi, and exact phrases and wildcards functionality
  - Single Words: This will match on the complete word. If you want to do a partial search, you will need to use wildcards.
  - Multiple Words: Use AND (e.g., `WordA AND WordB`) to find assets containing all terms, or OR to find one or the other.
  - Exact Phrases: Wrap your search in double quotes (e.g., `"search phrase"`).
  - Wildcards: Use * for partial matches. For example, `abc*` finds words starting with those letters, while `*abc*` finds words containing them, and `*abc` finds words ending in those letters.
  - Smart Matching: The system uses “stemming” to find related words. Searching for “crash” will automatically include results for “crashing” or “crashed”.
  - Case Sensitivity: Search is generally case-insensitive but operators like AND and OR must be capitalized.
- Aggregated data reflects only the matching hardware assets.

- As an IT Pro, I want to add filters to narrow the software asset list so I can build more precise queries.
  - Hardware data fields that can be used as Filters are defined in the Asset Data Spreadsheet
  - Software data fields that can be used as Filters are defined in the Asset Data Spreadsheet
  - Additional filter options
    - Scanner
    - Organization
  - Aggregated data reflects only the matching hardware assets.
- As an IT Pro, I want to be able to bookmark a specific filter so that I can reopen it later
  - This is different from Saved Filters.

Saving Searches

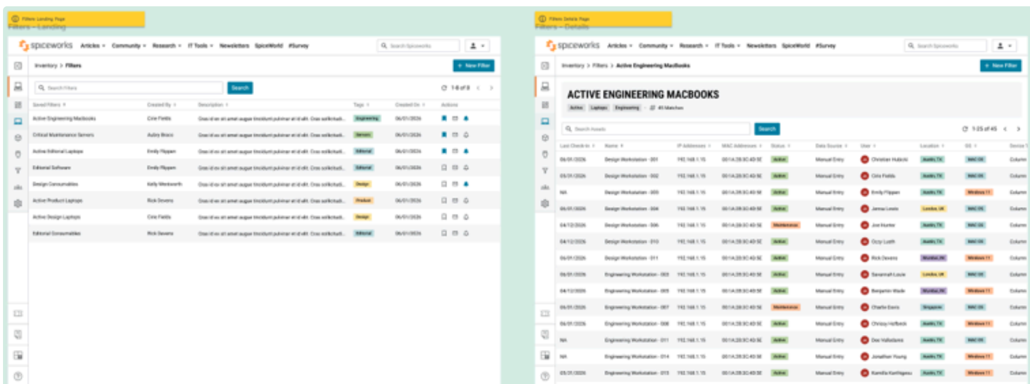
User Stories

- As an IT Pro, I want to save a search query or set of filters as a named view on either the Hardware or Software grid, so that I can quickly re-run common queries without rebuilding them.
  - When saving, the following information is required:

- Name
- Description
- Scope - Private or Public
- Filter Criteria / Query Parameters (system-populated)
- Created By (system-populated)
- DateTime Created (system-populated)

Reporting and Exports

See outcome requirements for a similar concept for CHD [Outcome] Reporting Improvements | In Review



Concept Design

Utilizing Saved Filters as Reports

- As an IT Pro, I want Saved Filters to serve as the mechanism for generating custom reports, so I can produce repeatable reports.
  - This applies to both software and hardware filters.
  - All users can see public Saved Filters
  - User can only see their own private Saved Filters
- As an IT Pro, I want to be able to see a list of Saved Filters for the account
  - Default Sort order: Favorites then Name

Managing Filters

User Stories

- As an IT Pro, I want to select a Saved Filter and see the matching assets displayed, so I can quickly access my commonly used views.
  - If a Saved Filter contains a custom field filter that has been deleted, that filter should be ignored.

- As an IT Pro, I want to update the search/filter criteria, name, description, and scope of an existing saved filter, or save it as a new filter, so that I can refine my views over time.
- As an IT Pro, I want to delete Saved Filters I no longer need to keep the filter list clean and relevant.
- As an IT Pro, I want to be able to mark specific filters as favorites so that I can easily find them

Default Out-of-the-Box Filters

User Stories

- As an IT Pro, I want pre-loaded default Saved Filters available on first login, so that I have immediate utility without needing to build views from scratch.
  - Default Saved Filters cannot be deleted.
  - All Hardware Assets: Shows every active hardware asset in the DB.
  - All Software Assets: Shows the global software inventory list.
  - Hardware Devices with Agents: Filters to assets where an active Collection Agent is registered.
  - Potential Duplicate Hardware Devices: Filters to assets flagged a potential duplicate.
  - Dynamic IP Agent Scopes: A default filter is automatically generated for each newly registered IP Scanning Agent, isolating only the assets discovered by that specific agent.

Exports

User Stories

- As an IT Manager, I want to export data from any Saved Filter or from the main Hardware and Software Asset Grids so I can use it for reports, in external tools, or share it with stakeholders.
  - The export should include all columns currently visible in the grid (respecting the column picker).
  - If Saved Filters or search are active, the export should include only the filtered/matching results.
- As an IT Manager, I want to export hardware asset data in CSV or XLSX format so I can choose the format that best suits my reporting needs.
- As an IT Manager, I want to export software asset data in CSV or XLSX format so I can choose the format that best suits my reporting needs.

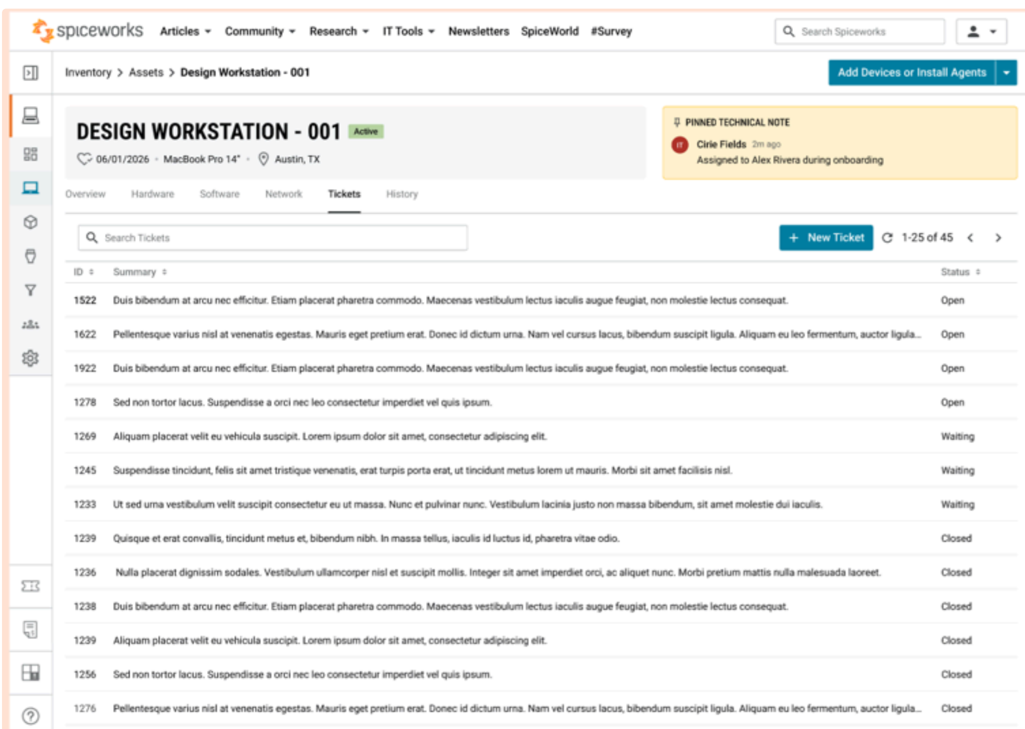
- As an IT Manager, I want to schedule exports from Saved Filters so I don’t have to remember to run them manually each time.
    - See outcome requirements for a similar concept for CHD [Outcome] Reporting Improvements | In Review
- As an IT Manager, I want to export all asset data from my account in a readable format.

Cloud Help Desk (CHD) Integration

The Inventory application must operate seamlessly alongside the Spiceworks Cloud Help Desk to enrich the ticketing context.

To be determined: How CHD and Inventory accounts are linked, dependant on the decision over the Freemium model.

Inventory



Concept Design

User Stories

- As an IT Technician, I want a "Create Support Ticket" button in the Asset Detail View that launches the CHD ticket creation dialog with the asset's reference pre-populated and locked, so that I can quickly raise tickets against a specific asset without re-entering details.
    - Supports the same create ticket flow as within CHD
    - Any fields required for a new ticket within CHD are also required within Inventory

- As an IT Technician, I want a dedicated "Tickets" tab in the Asset Detail View that shows all historical and active tickets associated with the device, so I can see the full support history for any asset.
  - Users cannot manage tickets from within Inventory.
  - Clicking a Ticket ID or row should open the ticket in CHD, not within Inventory.
- As an IT Manager, I want to see the Mean Time Between Failures (MTBF) for an asset, so that I can identify unreliable hardware and make informed replacement decisions.
  - Calculation: The average time between ticket creation dates for tickets associated with this asset. If the asset has 0 or 1 tickets, MTBF should display "N/A" or "Insufficient data"
    - Display in a human-readable string (e.g., "45 days")
- As an MSP, I want inventory assets to be automatically mapped to the correct CHD organization, so that tickets created from inventory are routed to the right client organization without manual selection.
  - When an MSP manages multiple client sites under a single account, each site's assets should be associated with the corresponding CHD organization.
  - When creating a ticket from the Asset Detail View, the CHD organization should be pre-populated based on the asset's site/organization mapping.
  - If an asset's site/organization has no corresponding CHD organization, the organization field in the ticket creation dialog should be left blank (not defaulted to a random organization), and the user should be prompted to select one manually.
  - A configuration interface should allow administrators to map Inventory sites/organizations to CHD organizations. This mapping may not be 1:1 — one CHD organization could map to multiple Inventory sites, or vice versa.
- As an IT Pro who does not use Cloud Help Desk, I want the Inventory application to function fully without CHD, so that I'm not blocked by features that require a product I don't have.
  - The "Create Support Ticket" button should be disabled with a hover-over message: *"Connect to Cloud Help Desk to create tickets from Inventory."*
  - The Tickets tab should display an empty state: *"Connect to Cloud Help Desk to see ticket history for this asset."*
  - The MTBF field should display "N/A - requires Cloud Help Desk."

Cloud Help Desk

User Stories

- As an IT Technician, I want the associated assets displayed on the CHD ticket to be linked to the asset's Inventory detail page, so that I can quickly access the asset context while working on a ticket.
- As an IT Technician, I want to link one or more inventory assets to a CHD ticket from within the individual ticket view, so that I can associate relevant hardware with a support request without having to navigate to the Inventory application first.
  - The technician should be able to search for and select assets from the inventory while working on a ticket in CHD. Search results should display enough context to identify the correct asset.
  - This is the reverse direction of the "Create Support Ticket" flow - it starts from CHD rather than from Inventory.
  - For MSPs, the search should default to the ticket's organization, with an option to search across all organizations.
- As an IT Technician, I want to remove an asset link from a CHD ticket to correct mistakes or remove irrelevant associations.
  - The unlink action should only be available from the CHD ticket view

User Access & Account Management

To be determined: How accounts are managed, especially in conjunction with CHD, if we opt for a Freemium Model.

User Stories

- As a product manager, I want users to create their Inventory account via the unified registration flow in ID Service to maintain a single registration process.
- As a product manager, I want users to log in via the unified login in ID Service to maintain a single login process.
- As an account administrator, I want to invite additional users to my existing Inventory account so my team can access and manage the inventory.
- As an account administrator, I want to view and manage which users have access to my Inventory account, so that I can remove users who no longer need access and maintain control over who can see and modify asset data.
- As an account administrator, I want users to be able to authenticate via Single Sign-On (SSO), so that my team can use their existing corporate identity provider and I can enforce centralized access policies.

- As a Spiceworks Technical Support team member, I need to be able to impersonate a user so I can see exactly what is happening on their account for troubleshooting.
- As the account owner, I want to be able to delete my account if it’s no longer needed.
- As a Spiceworks Technical Support team member, I want to be able to delete an account when the user requests it.

System Troubleshooting Log

User Stories

- As a Spiceworks Technical Support team member, I want a centralized, queryable log of all significant system events across all customer accounts, so that I can investigate and resolve customer issues, diagnose data discrepancies, and support debugging efforts without requiring engineering intervention.
    - This log is internal only - it is not exposed in the customer-facing UI and is not accessible to customers.
    - Accessible by Spiceworks internal support engineers via direct database querying
    - Each log entry should include at a minimum:
        - Timestamp (UTC)
        - Account/tenant identifier
        - User identifier (where applicable — or "System" for automated events)
        - Event type (categorized, so logs can be filtered by category)
        - Event detail (structured data relevant to the event — not free text)
        - Correlation ID or request ID (to trace a single user action across multiple system events)
- As a Spiceworks Technical Support team member, I want local logging for the deployed agent so that users can share this information with me when requesting assistant.
    - The existing app has 2 levels, normal/debug, which may be useful for avoiding excessive logging under normal operation but allowing a deep dive when a support ticket is open.

Out of Scope

- Migration of existing Cloud Inventory from existing accounts to the new Inventory application. Users will be required to create a new account and then install the new agents.

Non-Functional Requirements

Performance

- Initial asset discovery from a newly installed scanning or collection agent should begin returning results within hours, not days or weeks. The current Cloud Inventory experience of waiting days for initial scan data is not acceptable.
- Agent check-in processing should complete within a timeframe that allows the Hardware Asset Grid to reflect reasonably current data without noticeable lag.
- The Hardware Asset Grid and Software Asset Grid should load within acceptable performance expectations for list pages, including when an account has thousands of assets.
- Searching, sorting, filtering, and paginating the Hardware and Software Asset Grids should return results within acceptable performance expectations for list pages.
- The Asset Detail View should load within acceptable performance expectations, including tabs with multi-record grids (Storage, Network, Peripherals, Users, Software).
- Bulk update operations should not block the user from continuing to use the application. Large bulk updates should be processed as background operations.
- Saved Filter execution, including filters with complex criteria or large result sets, should return results within acceptable performance expectations for list pages.
- Export generation (CSV/XLSX) should follow acceptable performance expectations for file generation.
- Export generation must not materially degrade the performance of Inventory workflows for other users.

Scalability

- The system must support accounts ranging from a single device to thousands of assets without material degradation in grid performance, search response times, or agent check-in processing.
- The system must support MSP accounts managing multiple sites/organizations without performance degradation as the number of sites or total assets across sites increases.
- Agent check-in processing must scale to handle concurrent check-ins from many agents without throttling loops, blind retries, or the congestion patterns present in the current architecture.
- The system must support accounts with no defined limit on the number of Saved Filters, custom fields, or manual notes. Engineering should monitor and manage database performance, indexing, and query performance to ensure large accounts remain usable.

Agent Architecture

- Agents should send only changes (delta syncs) after the initial full scan, with an option to force a full re-sync. Full data uploads on every check-in are not acceptable.

- Registration traffic (new agents coming online) should be separated from update traffic (ongoing check-ins) to allow independent scaling and prioritization.
- Error logging from agents should not be auto-sent on every check-in. Errors should be pulled on demand or sent on a controlled schedule (e.g., weekly snapshot) to avoid contributing to congestion.
- Agents must implement appropriate backoff and retry behavior when the server is unreachable. Blind retries without randomization or exponential backoff are not acceptable.
- Agent updates should be delivered automatically, without requiring the user to manually reinstall.

Data Integrity

- Discovered fields must be updated accurately on each agent check-in. The system must handle partial or malformed agent payloads gracefully, accepting valid data and rejecting or logging invalid data without corrupting existing records.
- Manual and custom field values must never be overwritten by agent check-ins. The separation between discovered, calculated, manual, and custom field types must be enforced at the data layer, not just the UI.
- Bulk update operations must be atomic per asset; if a bulk update fails partway through, assets that were successfully updated should retain their new values, and assets that failed should retain their previous values. No asset should be left in a partially updated state.
- The audit history must accurately reflect all changes, including the correct old value, new value, user, timestamp, and change source. Audit entries must not be lost, duplicated, or attributed to the wrong user or change source.
- Install Count for software must remain consistent with the actual number of devices where that software is currently installed. Counts must be updated when software is detected, removed, or when assets are deleted or merged.
- Duplicate detection matching must be deterministic and consistent; the same data should always produce the same duplicate flags.

Security and Permissions

- There is no user role management for Inventory; all users who have access can perform all actions. However, access to the Inventory account itself must be enforced.
- The System Troubleshooting Log must not be accessible to customers. Access must be restricted to Spiceworks internal teams.
- Agent communication with the server must be encrypted in transit.

- Agent installers should be code-signed to reduce the likelihood of security software flagging them as malicious.

Account Isolation

- All asset data, software data, user data, custom fields, Saved Filters, exports, audit history, notes, agent registrations, and system logs must remain isolated to the correct Inventory account. No account should be able to access another account's data.
- For MSP accounts with multiple sites/organizations, data isolation between sites must be enforced at the query level, and site-scoped filters must not leak assets from other sites.

CHD Integration

- The CHD integration must not introduce a hard dependency. Inventory must function fully without a CHD account. CHD-dependent features (ticket creation, ticket history, MTBF) must degrade gracefully when CHD is not available.
- Asset-to-ticket associations must remain consistent across both systems. If an asset is deleted in Inventory, the historical ticket association in CHD must be preserved.
- Data freshness between Inventory and CHD should be sufficient that the Tickets tab in the Asset Detail View reflects reasonably current ticket data without requiring manual refresh.

Reliability

- Agent check-in processing must survive deployments, service restarts, and failovers without losing, duplicating, or corrupting incoming agent data.
- Saved Filter subscriptions must survive deployments and service restarts without being lost, duplicated, or run unexpectedly.
- If an agent check-in fails to process, the failure should be logged with sufficient detail for support to diagnose the issue. The agent should retry on its next scheduled check-in without requiring manual intervention.

Observability

- The system must provide sufficient database records and logs for support and engineering teams to troubleshoot agent registration, agent check-ins, asset creation, asset updates, bulk operations, merges, deletions, duplicate detection, Saved Filter execution, export generation, and CHD integration issues.
- The System Troubleshooting Log must capture all events defined in the functional requirements, with structured data queryable by account ID, user ID, event type, date range, asset ID, and agent ID.

- Relevant Datadog logs added for all actions / API calls with the correct level to aid troubleshooting and debugging.
- Appropriate alerts should be configured for degraded behavior, such as agent check-in processing backlogs, repeated check-in failures, export generation failures, or abnormal spikes in error rates.

Browser and Device Compatibility

- The Inventory web application must work in the browsers and device types currently supported by Spiceworks Cloud products.

Cost

- Operational costs for the new Inventory infrastructure (compute, storage, data transfer, agent processing) must be considered during architecture design.
- Storage costs for audit history (retained indefinitely), manual notes (with file attachments), and software change history should be estimated during engineering architecture to ensure long-term sustainability.

Post Launch Review

Launch Date

Useful reports

2 weeks

4 weeks

3 months

Questions and Decisions

Question Log

Date	Question	Who	Answer

Decision Log

Date	Decision	Rationale	Decision Makers

Links and Additional Documents

Engineering Research

Links to engineering research documentation.

User Interviews

Arthur Munoz: https://docs.google.com/document/d/11HCPkC2DwQjCo5VUW107Ycn5VBLyF2VciLwjww06-_o/edit?tab=t.ip8hzlkrev71 Can't find link

Community User 1 https://docs.google.com/document/d/114xLOVDrAxpiPhNi5jMKgbryYBww6ZMTTQxyoLnqPy0/edit?tab=t.qzojq67lsesr#heading=h.sivpaatrhr3m Can't find link

Community User 2 https://docs.google.com/document/d/1aoDYe55zEJwt0uowTzV-joGHoJxNMpK9xO9a9CWI-Ig/edit?tab=t.1hdn1a51ukhz Can't find link

ZDPM https://docs.google.com/document/d/15tl42iNt7TR982c9OMif-TImIz-5HEwIQ1dad0OVe7c/edit?tab=t.p7iev0quzb0m Can't find link

  Requirements list from ZDPM team https://docs.google.com/document/d/1-EStc7zqVXdljTyuPd4AFxZgkb6JooEZ/edit Can't find link

Meetings

https://docs.google.com/document/d/1NgFMFO_hEBHtnlFnjuwYu3N4CYtamDDrKbKNTF2sIgo/edit?tab=t.qh7jordlap5i Can't find link

https://docs.google.com/document/d/1k3MFJxSydODw8tDnddNmvsOyUygVEFhw892tkxKVE38/edit?tab=t.h2gplwtsrotj Can't find link

Inventory working session - 2026/04/28 15:00 BST - Notes by Gemini

https://docs.google.com/document/d/1OGVu5ExnwRZSCuv5qV8g6xN1XoinI7CZQdFXqpBoqTg/edit?tab=t.itiv0c7q3klj Can't find link

Inventory working session - 2026/04/14 14:58 BST - Notes by Gemini https://docs.google.com/document/d/1jGkjaDORwzSowxzlAyFbMPPRXQOVd-XQOaRIo-y9RiY/edit?tab=t.uprk6lipqd2v Can't find link

https://docs.google.com/document/d/1jGkjaDORwzSowxzlAyFbMPPRXQOVd-XQOaRIo-y9RiY/edit?tab=t.1ijptxnzyxb3 Can't find link

Inventory working session - 2026/03/31 14:59 BST - Notes by Gemini

Inventory working session - 2026/03/17 15:00 GMT - Notes by Gemini

Other Documents

Documentation: Inventory Online - Spiceworks

Inventory Online Scan Details - Spiceworks

Community Inventory Topics datapull (Apr 22, 2026)

Spiceworld 2019 IT Tools Session (includes a walk-through of Inventory) Doing IT Better With the Latest Spiceworks Tools and Apps

Spiceworks Dekstop App Video Spiceworks: Inventory your whole network